# Past Paper 02

Erik An

*<2026-02-16 Mon>*

## Contents

1. Is the set $\mathbb{R}^2$ with the component-wise defined operations a field?

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 y_1 \\ x_2 y_2 \end{pmatrix}$$

2. Let the field $\mathbb{Z}_p$ be given for $p$ prime, and consider the vector space $\mathbb{Z}_p^n$ for $n \geq 2$. For $0 \neq (v_1, \ldots, v_n) =: v \in \mathbb{Z}_p^n$ and $Q := (Q_1, \ldots, Q_n) \in \mathbb{Z}_p^n$, we define the line $g_{Q,v}$ in the direction $v$ through $Q$ as the set:

$$g_{Q,v} := \{Q + rv \mid r \in \mathbb{Z}_p\}$$

Determine the set of all lines in the vector spaces $\mathbb{Z}_2^3$ and $\mathbb{Z}_3^2$. Make a sketch.

3. Let $V$ be a vector space over $K$ and $X$ be a non-empty set. Show that the set of functions $\mathcal{F}(X, V)$ from $X$ to $V$ constitutes a vector space under pointwise operations. Use different symbols for all occurring operations during the verification.

4. Are the following subsets of $\mathbb{R}^3$ subspaces (with justification)?
   (a) $\{(x, y, z) \mid y = 0\}$
   (b) $\{(x, y, z) \mid x = 1\}$
   (c) $\{(x, y, z) \mid x = 0 \text{ or } y = 0 \text{ or } z = 0\}$
   (d) $\{(x, y, z) \mid ax + by = cz\}$ for fixed $a, b, c \in \mathbb{R}$.
   (e) $\{(x, y, z) \mid x^2 + y^2 = z\}$

5. Let $K$ be a field. Show that the vector space $K$ over the field $K$ possesses only the trivial subspaces $\{0\}$ and $K$.

6. Let $V$ be a vector space over $K$ and $W$ a subspace of $V$.

   Then $W$ itself is again a vector space over $K$. Let $U$ be a subspace of $W$.

   Show that $U$ is then also a subspace of $V$.

7. Show directly that the functions $f, g \colon \mathbb{R} \to \mathbb{R}$, given by $f(x) := 3x + 5$ and $g(x) = x^2$, are not linear maps.

   Determine all linear maps $f \in L(\mathbb{R}, \mathbb{R})$.

8. Which of the following maps is linear?
   (a) $f_1 \colon \mathbb{R}^3 \to \mathbb{R}$, $(x, y, z) \mapsto 4x + 3y - 15z$
   (b) $f_2 \colon \mathbb{R}^3 \to \mathbb{R}$, $(x, y, z) \mapsto 2x + y - 2z + 5$
   (c) $f_3 \colon \mathbb{R}^3 \to \mathbb{R}^2$, $(x, y, z) \mapsto (x + y - z, 3x + 15z)$
   (d) $f_4 \colon \mathbb{Z}_2^3 \to \mathbb{Z}_2^2$, $(x, y, z) \mapsto (x^2 + 2xz + z^2 + x + z, x^2 + y^2 + z^2)$
   (e) $f_5 \colon \mathbb{C}^3 \to \mathbb{C}^2$, $(x, y, z) \mapsto (x^2 + 2xz + z^2 + x + z, x^2 + y^2 + z^2)$

9. Investigate the map $\mathbb{C} \to \mathbb{C}$, $z \mapsto \overline{z}$, which assigns the complex conjugate to a complex number.

   Show that this map is linear if $\mathbb{C}$ is viewed as a vector space over $\mathbb{R}$.

   Is this also the case if one views $\mathbb{C}$ as a vector space over $\mathbb{C}$?

   What is the situation if $\mathbb{C}$ is viewed as a vector space over $\mathbb{Q}$?

10. Let $\mathcal{F}_{\mathbb{R}} := \mathbb{R}^{\mathbb{N}} := \{(x_n)_n := (x_0, x_1, x_2, \dots) \mid x_i \in \mathbb{R}\}$ be the set of all real number sequences.

    Show that $\mathcal{F}_{\mathbb{R}}$ forms a vector space over $\mathbb{R}$ with component-wise addition and scalar multiplication:

$$(x_0, x_1, x_2, \dots) + (y_0, y_1, y_2, \dots) := (x_0 + y_0, x_1 + y_1, x_2 + y_2, \dots)$$

$$r \cdot (x_0, x_1, x_2, \dots) := (rx_0, rx_1, rx_2, \dots)$$

    for all $(x_n)_n, (y_n)_n$ and $r \in \mathbb{R}$, and investigate whether the subset $c$ of all convergent sequences forms a subspace.

    Let $c := \{(x_n)_n \mid \exists \lim_{n\to\infty} x_n = x \in \mathbb{R}\}$.

    Let $c_0$ be the set of all null sequences (sequences converging to 0).

    What holds for $c_0$ (regarding vector space properties)?

    Is the map $\lim\colon c \to \mathbb{R}$ with $(x_n)_n \mapsto \lim_{n\to\infty} x_n$ a linear map?

11. Analogous to Problem 10, consider the set of all sequences $\mathcal{F}_{\mathbb{K}} = \mathbb{K}^{\mathbb{K}}$ of elements of a field $K$.

    Justify why $\mathcal{F}_{\mathbb{K}}$ is a vector space. Show that the subset of all sequences $\mathbb{K}^{(\mathbb{N})}$, which have only finitely many non-zero terms, is a subspace.

$$\mathbb{K}^{(\mathbb{N})} := \{(x_n)_n \in \mathcal{F}_{\mathbb{K}} \mid \exists N \geq 0 : \forall k \geq N : x_k = 0\}$$

    Recall the definition of a polynomial over $K$ from the lecture.

    Clarify for yourself: A polynomial $p = \sum_{k=0}^{n} c_k X^k$ is uniquely determined by the (infinitely extended) sequence of its coefficients $p \sim (c_0, c_1, c_2, \dots)$ (here all sequence terms $c_k$ with $k > \deg p$ are equal to 0).

    Show that the vector space $\mathbb{K}[X]$ of polynomials over $K$ is isomorphic to $\mathbb{K}^{(\mathbb{N})}$.

12. From school, you know the multiplication of two real polynomials $p$ and $q$.

    Provide an operation corresponding to this product $p \cdot q$ that is defined on $\mathbb{K}^{(\mathbb{N})}$, i.e., uses only the coefficients and not the variable $X$.

    (Hint: The sought expression is, for each coefficient of the resulting sequence, a sum of products of elements from $K$.) Show that $\mathbb{K}^{(\mathbb{N})}$ forms an integral domain and a $K$ -Algebra with addition and the multiplication defined in this way.

13. Let $\mathbb{R}[X]$ be the vector space of polynomials with real coefficients.

    Show that the functions $D\colon \mathbb{R}[X] \to \mathbb{R}[X]$ with $D(\sum_k a_k X^k) = \sum_k k a_k X^{k-1}$ and $M\colon \mathbb{R}[X] \to \mathbb{R}[X]$ with $M(\sum_k a_k X^k) = \sum_k a_k X^{k+1}$ are linear, and determine $\mathrm{Ker}(D)$, $\mathrm{Ker}(M)$, $\mathrm{Im}(D)$, and $\mathrm{Im}(M)$.

    What kind of maps are these? Show that for all $n > 0$, $D \circ M^n - M^n \circ D = nM^{n-1}$ holds, where $M^0 := \mathrm{id}$ and $M^{k+1} := M \circ M^k$.

14. (a) Let $f\colon V \to W$ be a linear map between two $K$ -vector spaces.

Show that for any arbitrary further $K$ -vector space $U$, a linear map $f_*\colon L(U,V) \to L(U,W)$ is defined by $f_*(g) := f \circ g$.

(b) Let $f\colon V \to W$ be a linear map between two $K$ -vector spaces.

Show that for any arbitrary further $K$ -vector space $Z$, a linear map $f^*\colon L(W,Z) \to L(V,Z)$ is defined by $f^*(g) := g \circ f$.

15. (a) Determine the products $AB$, $BA$, and $(AB)^2 = ABAB$ for the following matrices $A$ and $B$ over $\mathbb{Z}_5$.

$$A = \begin{pmatrix} 3 & 2 & 4 \\ 3 & 2 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 4 & 4 \\ 4 & 3 \\ 2 & 4 \end{pmatrix}$$

Can one form $A^2 = AA$ or $B^2 = BB$?

(b) Calculate the products $Ax$ and $Ay$ for $A = \begin{pmatrix} 1 & i & 2-i & -3i \\ 0 & 3+2i & -1 & 1-2i \end{pmatrix} \in M_{2,4}(\mathbb{C})$ and $x = \begin{pmatrix} 1 \\ -i \\ -1 \\ 1+3i \end{pmatrix}$, $y = \begin{pmatrix} -2 \\ 1+i \\ 0 \\ i \end{pmatrix} \in \mathbb{C}^4$.

(c) Calculate $A^2 = AA$ and $A^3$ for $A = \begin{pmatrix} 2 & 3 \\ 3 & -2 \end{pmatrix} \in M_2(\mathbb{R})$. Does anything change if one views $A$ as an element of $M_2(\mathbb{Z}_{13})$?

16. (a) Find a $3 \times 3$ matrix $A$ over $\mathbb{R}$ such that $A$ and $A^2$ are not the null matrix, but $A^3$ is the null matrix.

Is this also possible for $2 \times 2$ matrices? Instruction: Look for a suitable linear map $\mathbb{R}^3 \to \mathbb{R}^3$.

(b) A matrix $N \in M_n(\mathbb{K})$ is called nilpotent if there exists a $k \in \mathbb{N}$ such that $N^k = 0$ (the null matrix).

Show that $I - N$ is then invertible and that:

$$(I-N)^{-1} = \sum_{j=0}^{k-1} N^j$$

17. For which $b_1, b_2, b_3 \in \mathbb{C}$ does the following linear system of equations possess a solution over $\mathbb{C}$?

$$\begin{aligned} 9x_1 - 7ix_2 + 8x_3 &= b_1 \\ -ix_1 - 2x_2 + x_3 &= b_2 \\ ix_1 - 8x_2 + (11+16i)x_3 &= b_3 \end{aligned}$$

Calculate the solution depending on the $b_i$.

18. Determine the set of all solutions of the following linear system of equations for $b_1, b_2 \in \mathbb{Z}_5$.

$$4x_1 + 3x_2 = b_1$$
$$2x_1 + 3x_2 = b_2$$

19. Determine the inverse matrix of the following over $\mathbb{Z}_7$.

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 5 \\ 1 & 0 & 5 \end{pmatrix}$$

20. Determine the kernel and image, as well as the matrix representation, of the linear map $f\colon \mathbb{R}^3 \to \mathbb{R}^3$ given by:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} 5x + 19y + 2z \\ x - 4y + z \\ 3x + y + 2z \end{pmatrix}$$

21. Investigate whether the set $\{(2,1,3),(1,1,0),(3,0,4)\}$ is a generating set for $\mathbb{R}^3$.

    Is it a generating set for $\mathbb{Q}^3$ or $\mathbb{C}^3$?

22. Show directly that the polynomials $B_0 = X^2$, $B_1 = -2X^2 + 2X$, and $B_2 = X^2 - 2X + 1$ are linearly independent in $\mathbb{R}[X]$.

    Show furthermore that they are a generating set for $\mathbb{R}_2[X]$. Calculate $B_0 + B_1 + B_2$.

23. Show that the set $\{X^k \mid k \in \mathbb{N}\}$ forms a basis for the vector space $\mathbb{K}[X]$ over an arbitrary field $K$.

    Conclude from this that $\mathbb{K}[X]$ is not finitely generated.

24. Show that the set $C^0(\mathbb{R},\mathbb{R})$ of continuous functions from $\mathbb{R}$ to $\mathbb{R}$ forms a vector space, and show that the set $\{e^{kx} \mid k \in \mathbb{Z}\}$ is linearly independent in $C^0(\mathbb{R},\mathbb{R})$.

25. Check whether the set $\{(0,1,1),(1,-1,0),(1,0,1)\}$ is linearly independent in $\mathbb{R}^3$.

    Is it linearly independent in $\mathbb{C}^3$, or in $\mathbb{Z}_2^3$?

26. Let $V$ be a $K$ -vector space, and let $\{W_i \mid i \in I\}$ be a non-empty family of subspaces of $V$, such that for all pairs $(i,j) \in I^2$ there exists a $k \in I$ with $W_i \cup W_j \subseteq W_k$.

    Show that $\bigcup_{i \in I} W_i$ is a subspace of $V$. What about the converse of this statement?

27. Let $V$ be an $n$ -dimensional vector space over a field $K$ and $W$ a subspace of dimension $n-1$.

    Show:

    (a) There exists a linear map $f\colon V \to \mathbb{K}$ such that $W = \operatorname{Ker}(f)$.

    (b) If $g\colon V \to \mathbb{K}$ is another linear map with $\operatorname{Ker}(g) = W$, then there exists an element $r \in \mathbb{K} \setminus \{0\}$ such that $g(v) = rf(v)$ for all $v \in V$.

    Instruction for (a): Extend a basis for $W$ to a basis for $V$ and define $f$ on the elements of this basis.

28. Let $V$ be an $n$ -dimensional vector space over a field $K$ and $W$ a $k$ -dimensional subspace of $V$.

    Show:

    (a) There exists a (surjective) linear map $f\colon V \to \mathbb{K}^{n-k}$ such that $W = \operatorname{Ker}(f)$.

    (b) If $g\colon V \to \mathbb{K}^{n-k}$ is another linear map with $W = \operatorname{Ker}(g)$, then there exists a (unique) linear isomorphism $\varphi\colon \mathbb{K}^{n-k} \to \mathbb{K}^{n-k}$ such that $g = \varphi \circ f$.

    (c) There exists an (injective) linear map $f\colon \mathbb{K}^k \to V$ such that $W = \operatorname{Im}(f)$.

    (d) If $g\colon \mathbb{K}^k \to V$ is another linear map with $W = \operatorname{Im}(g)$, then there exists a (unique) linear isomorphism $\varphi\colon \mathbb{K}^k \to \mathbb{K}^k$ such that $g = f \circ \varphi$.

29. (a) Determine a complement to the plane $\{(x, y, z) \mid 3x + 4y - z = 0\}$ in $\mathbb{R}^3$.

    (b) Show that every line $g_b := \{(t, tb, 0) \mid t \in \mathbb{R}\}$ with $b \in \mathbb{R} \setminus \{0\}$ is a complement for the plane $\{(x, 0, z) \mid x, z \in \mathbb{R}\}$.

30. Let $n \in \mathbb{N}$ and $p \in \mathbb{R}[X]$ with $\deg(p) = n + 1$. Let $p \cdot \mathbb{R}[X]$ denote the set of all polynomials divisible by $p$.

    Show:

    (a) $p \cdot \mathbb{R}[X]$ is a subspace of $\mathbb{R}[X]$.

    (b) $\mathbb{R}_n[X]$ and $p \cdot \mathbb{R}[X]$ are complementary in $\mathbb{R}[X]$.

31. Show that the sets of even functions $\mathcal{F}_g(\mathbb{R}, \mathbb{R}) := \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid \forall x \in \mathbb{R} : f(x) = f(-x)\}$ and of odd functions $\mathcal{F}_u(\mathbb{R}, \mathbb{R}) := \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid \forall x \in \mathbb{R} : f(x) = -f(-x)\}$ are complementary in $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

    Do not forget to show that $\mathcal{F}_g(\mathbb{R}, \mathbb{R})$ and $\mathcal{F}_u(\mathbb{R}, \mathbb{R})$ are subspaces.

32. Let $V$ be a finitely generated $K$ -vector space, $W \subset V$ a subspace.

    Show that there exists a projection onto $W$, i.e., a linear map $\pi\colon V \to W$, such that $\pi(w) = w$ for all $w \in W$.

    Show furthermore that $\operatorname{Ker}(\pi) \subset V$ is a complement to $W$. Can every complement be obtained this way?

33. Determine the rank of the matrix over $\mathbb{R}$.

$$\begin{pmatrix} 3 & 4 & 1 & 8 & 9 \\ -2 & 0 & 2 & 3 & 0 \\ 1 & 2 & -1 & 1 & 2 \\ 7 & 8 & -3 & 7 & 13 \end{pmatrix}$$

    What is the rank over $\mathbb{Z}_{17}$?

34. (a) Consider the standard basis $\mathcal{S}$ and the basis $\mathcal{B} := \{(1, 3), (2, 2)\}$ of $\mathbb{Z}_5^2$.

    Calculate the change-of-basis maps from $\mathcal{S}$ to $\mathcal{B}$ and vice versa.

    Determine the coordinate vector $[v]_{\mathcal{B}}$ for the vector $v = (2, 4)$ (relative to the standard basis).

    For the linear map $f\colon \mathbb{Z}_5^2 \to \mathbb{Z}_5^2$ with $(x, y) \mapsto (3x + 2y, 4x + y)$, determine the matrix representations $[f]_{\mathcal{S}}^{\mathcal{B}}$ and $[f]_{\mathcal{B}}^{\mathcal{B}}$.

(b) Consider the standard basis $\mathcal{S}$ and the bases $\mathcal{B} := \{(1,1,0),(0,-1,1),(-1,0,1)\}$ and $\tilde{\mathcal{B}} := \{(2,0,1),(1,1,1),(-1,3,0)\}$ of $\mathbb{R}^3$.

Determine the change-of-basis maps from $\mathcal{B}$ to $\tilde{\mathcal{B}}$, from $\mathcal{B}$ to $\mathcal{S}$, and from $\tilde{\mathcal{B}}$ to $\mathcal{S}$.

Determine the coordinate vectors $[v]_{\mathcal{B}}$ and $[v]_{\tilde{\mathcal{B}}}$ for the vector $v = (1,4,2)$ (relative to the standard basis).

For the linear map $f\colon \mathbb{R}^3 \to \mathbb{R}^3$ with $(x,y,z) \mapsto (3x+2z, x-4y, 6y-z)$, determine the matrix representations $[f]_{\mathcal{S}}^{\mathcal{S}}$, $[f]_{\mathcal{S}}^{\tilde{\mathcal{B}}}$, and $[f]_{\mathcal{B}}^{\tilde{\mathcal{B}}}$.

35. Show that there exist real numbers $r_0, r_1, r_2 \in \mathbb{R}$ such that for every polynomial $p \in \mathbb{R}_2[X]$, the associated polynomial function $p\colon \mathbb{R} \to \mathbb{R}$ satisfies the equation $p(-1) = r_0 p(0) + r_1 p(1) + r_2 p(2)$.

    Determine these numbers explicitly.

36. Show that there exist real numbers $s_0, s_1, s_2 \in \mathbb{R}$ such that for every polynomial $p \in \mathbb{R}_2[X]$, the associated polynomial function $p\colon \mathbb{R} \to \mathbb{R}$ satisfies the equation:

$$D(p)(0) = s_0 p(-2) + s_1 p(0) + s_2 p(2)$$

    where $D\colon \mathbb{R}_2[X] \to \mathbb{R}_2[X]$ is the (formal) derivative operator on the polynomials.

37. Consider the vector space $\mathbb{R}_3[X]$. Show that the polynomials $B_{3,i} := \binom{3}{i} X^i (1-X)^{3-i}$ for $i = 0, \ldots, 3$ form a basis $\mathcal{B}$.

    Determine the change-of-basis maps $[\mathrm{id}]_{\mathcal{B}}^{\mathcal{S}}$ and $[\mathrm{id}]_{\mathcal{S}}^{\mathcal{B}}$, and calculate $[1]_{\mathcal{B}}$ and $[D]_{\mathcal{B}}^{\mathcal{B}}$, where $D\colon \mathbb{R}_3[X] \to \mathbb{R}_3[X]$ denotes the (formal) derivative operator and $\mathcal{S} := \{1, X, X^2, X^3\}$ the standard basis.

38. Consider the matrix:

$$A = \begin{pmatrix} 4 & 2 & 4 \\ 6 & 3 & 6 \end{pmatrix}$$

    Find bases $\mathcal{B}$ and $\mathcal{B}'$ such that $[A]_{\mathcal{B}'}^{\mathcal{B}} = \mathbb{J}_r$ for a suitable $r$ (see Theorem 4.18).

39. Consider the matrix:

$$A = \begin{pmatrix} 1 & 3 & 2 & -1 \\ -1 & -1 & 4 & 2 \\ 1 & 5 & 8 & 0 \end{pmatrix}$$

    Find bases $\mathcal{B}$ and $\mathcal{B}'$ such that $[A]_{\mathcal{B}'}^{\mathcal{B}} = \mathbb{J}_r$ for a suitable $r$ (see Theorem 4.18).

40. Let $v_1 = (1,1,-1)$, $v_2 = (1,2,4)$, $v_3 = (3,-1,r)$, $v_4 = (2,3,s)$ be vectors in $\mathbb{R}^3$. Determine $r, s \in \mathbb{R}$ such that $\langle\{v_1, v_2\}\rangle = \langle\{v_3, v_4\}\rangle$.

41. Let $V$ be a $K$ -vector space. We call a linear map $\pi\colon V \to V$ a projection if $\pi \circ \pi = \pi$.

    Let $\pi$ be a projection. Show that

    (a) $\mathrm{id} - \pi$ is also a projection,

(b) $\mathrm{Ker}(\pi) = \mathrm{Im}(\mathrm{id} - \pi)$,

(c) $\mathrm{Im}(\pi) = \mathrm{Ker}(\mathrm{id} - \pi)$,

(d) $V = \mathrm{Ker}(\pi) \oplus \mathrm{Im}(\pi)$.

42. Consider the subspace $W := \{x \in \mathbb{R}^3 \mid 2x_1 + 3x_2 - 2x_3 = 0\}$ of $\mathbb{R}^3$. Describe the cosets $v + W$ and investigate the vector space operations of $\mathbb{R}^3/W$, i.e., describe the arithmetic operations as explicitly as possible.

43. Consider the subspace $W := \langle\{(1,0,1,0),(1,1,0,1)\}\rangle$ of $\mathbb{Z}_2^4$. Describe the cosets $v + W$ and investigate the vector space operations of $\mathbb{Z}_2^4/W$.

44. Determine the dual basis of $\mathbb{R}^{3^*}$ for the basis $\mathcal{B} := \{(1,2,1),(-1,0,2),(1,-1,0)^T\} \subseteq \mathbb{R}^3$ and determine the dual map $f^*$ for the map $f\colon \mathbb{R}^3 \to \mathbb{R}^3$, $(x_1,x_2,x_3) \mapsto (3x_1 - x_2 + 2x_3, -x_1 + 2x_2 - x_3, 5x_1 + 4x_3)$.

45. Let $U, V$ and $W$ be vector spaces over $K$, and let $f \in L(U,V)$ and $g \in L(V,W)$.

    Show that

    (a) $\mathrm{Ker}(g \circ f) = f^{-1}(\mathrm{Ker}(g))$

    (b) $\mathrm{Ker}(g \circ f) \supseteq \mathrm{Ker}(f)$ with equality if and only if $\mathrm{Ker}(g) \cap \mathrm{Im}(f) = \{0\}$.

    (c) $\mathrm{Im}(g \circ f) \subseteq \mathrm{Im}(g)$ with equality if and only if $\mathrm{Ker}(g) + \mathrm{Im}(f) = V$.

46. Show that the sets $F := \{f \in \mathcal{F}(\mathbb{R},\mathbb{R}) \mid f(1) = 0\}$ and $G := \{f \in \mathcal{F}(\mathbb{R},\mathbb{R}) \mid \exists a \in \mathbb{R} : \forall x \in \mathbb{R} : f(x) = ax\}$ are complementary subspaces of $\mathcal{F}(\mathbb{R},\mathbb{R})$.

47. Let $f\colon \mathbb{C}[X] \to \mathbb{C}[X]$ be the map $p \mapsto p(X+1) - p(X)$. Show that $f$ is linear and determine $\mathrm{Ker}(f)$. Is the map surjective?

    Show furthermore that there exists a basis $(b_0, b_1, \dots)$ of $\mathbb{C}[X]$ such that $f(b_{i+1}) = b_i$ for all $i \in \mathbb{N}$.

48. Let $V$ be a $K$ -vector space and $p, q\colon V \to V$ two projections (i.e. $p \circ p = p$).

    Show that $p + q$ is a projection if and only if $p \circ q = q \circ p = 0$.

49. Let the vector space $\mathbb{K}_n[X]$ be given, and let $x_0, x_1, \dots, x_n$ be distinct elements of $K$.

    Show that the evaluations $\mathrm{ev}_{x_i}\colon \mathbb{K}_n[X] \to \mathbb{K}$, $p \mapsto p(x_i)$ for $i = 0, \dots, n$ form a basis for $\mathbb{K}_n[X]^*$.

    For which basis is $\mathrm{ev}_{x_0}, \dots, \mathrm{ev}_{x_n}$ the dual basis?

50. Prove the Theorem of Menelaus: Let three distinct points $A, B$, and $C$ be given in $\mathbb{A}^2$, forming a non-degenerate triangle.

    Let $g$ be a line passing through no vertex of the triangle and let $\{D\} = g \cap l(A,B)$, where $l(A,B)$ denotes the line passing through $A$ and $B$.

    Let $\{E\} = g \cap l(B,C)$ and $\{F\} = g \cap l(C,A)$. Then:

$$\frac{A-D}{B-D}\frac{B-E}{C-E}\frac{C-F}{A-F} = 1$$

51. Prove the Theorem of Ceva: Let four distinct points $A, B, C$, and $P$ be given in $\mathbb{A}^2$ such that $ABC$ is a non-degenerate triangle.

    Let $\{D\} = l(A, B) \cap l(C, P)$, $\{E\} = l(B, C) \cap l(A, P)$, and $\{F\} = l(C, A) \cap l(B, P)$. Under the assumption that all seven points are distinct:

$$\frac{A-D}{B-D}\frac{B-E}{C-E}\frac{C-F}{A-F} = -1$$